Set 1 (Questions 1–50) Topic wise

Importance of Biochemistry in Dentistry & Intro to Biochem

Q1

A first-year BDS student at Dow Dental College asks why molecular biology is integrated into the dental curriculum. According to the standard introductory concepts, biochemistry is essential because:

a. It teaches the manual dexterity required for clinical fillings

b. It helps with the physical architectural design of dental clinics

c. It is primarily used to study the macro-anatomy of the mandible

d. It improves patient comfort specifically during high-speed drilling

e. It explains molecular changes in the oral cavity that are common to other tissues

Q2

During an introductory lecture on the chemical logic of living organisms, the professor describes the "universal currency" of free energy. Which of the following molecules fulfills this role?

a. ATP

b. Glucose

c. NADH

d. FADH2

e. Cyclic AMP

Biochemical Composition & Functional Organization of the Cell

Q3

A researcher is studying a cellular malfunction where the detoxification of reactive oxygen species (free radicals) is impaired. Which single-membrane bound organelle is most likely the site of this defect?

a. Lysosome

b. Peroxisome

c. Rough Endoplasmic Reticulum

d. Smooth Endoplasmic Reticulum

e. Nucleolus

Q4

A patient presents with a genetic condition that weakens the "network of filaments" responsible for maintaining cell shape and internal organization. This defect primarily involves the:

a. Lipid bilayer

b. Cytoskeleton

c. Mitochondrial matrix

d. Golgi apparatus

e. Nuclear envelope

Q5

During the post-translational modification of proteins, which organelle serves as the primary site for the packaging, sorting, and formation of lysosomes?

a. Smooth Endoplasmic Reticulum

b. Nucleus

c. Mitochondria

d. Golgi Apparatus

e. Peroxisome

Water, pH & Buffers

Q6

Water is uniquely suited as a biological solvent due to its dipole nature and ability to form specific bonds. Which property allows water to have a high boiling point and high surface tension?

a. Hydrogen bonding

b. Ionic bonding

c. Van der Waals forces

d. Hydrophobic interactions

e. Covalent bonding

Q7

A 40-year-old male is brought to the emergency department with severe diarrhea. To maintain his physiological pH, his body utilizes the most significant extracellular buffer. This buffer is:

a. Bicarbonate

b. Hemoglobin

c. Phosphate

d. Acetate

e. Ammonium

Q8

A laboratory technician notes that the pH of a solution has changed from 5.0 to 4.0. This represents which of the following changes in the hydrogen ion [H+] concentration?

a. 2-fold decrease

b. 10-fold decrease

c. 20-fold increase

d. 100-fold increase

e. 10-fold increase

Q9

According to the Henderson-Hasselbalch equation, when the pH of a solution is equal to the pKa of the weak acid, the ratio of conjugate base to acid is:

a. 1:1

b. 10:1

c. 1:10

d. 100:1

e. 0:1

Q10

While the bicarbonate system is the major plasma buffer, which system acts as the primary buffer within the renal tubular fluid and the intracellular compartment?

a. Hemoglobin buffer

b. Carbonic acid buffer

c. Ammonia buffer

d. Phosphate buffer

e. Plasma protein buffer

Cell Membrane & Its Functions

Q11

The "Fluid Mosaic Model" describes the cell membrane. Which specific lipid component is most abundant in the bilayer and serves as the primary structural barrier?

a. Phospholipid

b. Cholesterol

c. Triglyceride

d. Sphingomyelin

e. Glycolipid

Q12

A 19-year-old boy has a total body water of 42L. If his extracellular fluid (ECF) is 14L and his plasma volume is 3L, what is his calculated interstitial fluid volume?

a. 4L

b. 6L

c. 11L

d. 9L

e. 16L

Extracellular Matrix (ECM) in Health & Disease

Q13

A patient with bleeding gums and poor wound healing is diagnosed with Scurvy. This condition prevents the post-translational hydroxylation of which amino acids?

a. Methionine and Cysteine

b. Glycine and Alanine

c. Proline and Lysine

d. Leucine and Isoleucine

e. Valine and Serine

Q14

A neonate presents with blue sclera and multiple fractures. The diagnosis is Osteogenesis Imperfecta. This disease is fundamentally a defect in the synthesis of:

a. Elastin

b. Fibronectin

c. Laminin

d. Collagen

e. Keratin

Q15

Which of the following glycosaminoglycans (GAGs) is unique because it is non-sulfated and is found abundantly in the vitreous humor and umbilical cord?

a. Chondroitin sulfate

b. Heparan sulfate

c. Keratan sulfate

d. Dermatan sulfate

e. Hyaluronic acid

Classification & Functions of Proteins

Q16

Proteins can be classified based on their composition. A protein that consists of a polypeptide chain combined with a non-protein prosthetic group (like Heme) is called a:

a. Simple protein

b. Derived protein

c. Incomplete protein

d. Conjugated protein

e. Fibrous protein

Q17

Which of the following proteins serves a vital protective role in the blood by acting as a major inhibitor of proteolytic enzymes like elastase?

a. Albumin

b. Alpha-1 antitrypsin

c. Transferrin

d. Ceruloplasmin

e. Haptoglobin

Q18

Structural proteins like Keratin and Collagen are characterized by being water-insoluble and having a length-to-width ratio greater than 10. These are classified as:

a. Globular proteins

b. Fibrous proteins

c. Catalytic proteins

d. Regulatory proteins

e. Transport proteins

Structure & Classification of Amino Acids

Q19

Which amino acid is classified as "non-essential" because it can be synthesized from 3-phosphoglycerate, a metabolic intermediate of glycolysis?

a. Valine

b. Lysine

c. Serine

d. Histidine

e. Phenylalanine

Q20

A child is diagnosed with Phenylketonuria (PKU). This disease results from a deficiency in an enzyme that converts Phenylalanine into which of the following?

a. Tryptophan

b. Tyrosine

c. Histidine

d. Methionine

e. Arginine

Q21

Which amino acid lacks an asymmetric alpha-carbon atom and is therefore the only achiral amino acid found in human proteins?

a. Alanine

b. Glycine

c. Valine

d. Leucine

e. Isoleucine

Q22

The "aromatic" side chain of which amino acid is a precursor for the synthesis of the neurotransmitter Serotonin and the vitamin Niacin?

a. Phenylalanine

b. Tyrosine

c. Tryptophan

d. Histidine

e. Proline

Q23

Which amino acid contains a thiol (-SH) group and is capable of forming covalent disulfide bridges to stabilize protein tertiary structure?

a. Methionine

b. Cysteine

c. Serine

d. Threonine

e. Glutamate

Structural Organization of Proteins

Q24

The "Primary Structure" of a protein refers to the linear sequence of amino acids. This structure is stabilized exclusively by which type of bond?

a. Hydrogen bond

b. Ionic bond

c. Disulfide bond

d. Peptide bond

e. Hydrophobic interaction

Q25

Which level of protein organization is characterized by local folding into patterns such as the alpha-helix or beta-pleated sheet?

a. Primary

b. Secondary

c. Tertiary

d. Quaternary

e. Domain

Q26

In a globular protein, which interaction is the primary driving force for folding, where nonpolar side chains are buried in the interior?

a. Hydrogen bonding

b. Electrostatic attraction

c. Hydrophobic effect

d. Covalent cross-linking

e. Van der Waals forces

Q27

A protein consisting of multiple polypeptide chains (subunits) that must come together to be functional (like Hemoglobin) is at which level of structure?

a. Primary

b. Secondary

c. Tertiary

d. Quaternary

e. Motif

Q28

Denaturation involves the loss of secondary, tertiary, and quaternary structure. Which of the following can cause protein denaturation?

a. Organic solvents

b. Strong acids or bases

c. Heat

d. Detergents

e. All of the above

Role of Proteins in Health & Disease (Hb & Mb)

Q29

The "Bohr Effect" describes the physiological shift in the oxygen dissociation curve of Hemoglobin. A shift to the right occurs in response to:

a. Increased pH

b. Decreased pCO_2

c. Decreased temperature

d. Increased H^+ concentration

e. Decreased 2,3-BPG

Q30

Fetal Hemoglobin (HbF) has a higher affinity for oxygen than Adult Hemoglobin (HbA). This is because HbF binds which of the following less strongly?

a. Oxygen

b. Carbon dioxide

c. 2,3-BPG

d. Protons (H^+)

e. Carbon monoxide

Q31

Sickle Cell Anemia is caused by a point mutation in the beta-globin chain. This mutation results in the substitution of:

a. Glutamate for Valine

b. Valine for Glutamate

c. Lysine for Glutamate

d. Glycine for Proline

e. Histidine for Tyrosine

Q32

Myoglobin has a hyperbolic oxygen dissociation curve, whereas Hemoglobin's curve is sigmoidal. The sigmoidal shape of Hemoglobin's curve indicates:

a. Lack of cooperativity

b. Cooperative binding

c. Irreversible binding

d. Low oxygen affinity at all pressures

e. Competition for the active site

Q33

Methemoglobinemia is a condition where the iron in heme is oxidized and cannot bind oxygen. What is the oxidation state of iron in Methemoglobin?

a. Fe0

b. Fe+1

c. Fe+2

d. Fe+3

e. Fe+4

Plasma Proteins & Immunoglobulins

Q34

Which plasma protein is responsible for transporting iron in the blood and typically increases in concentration during iron deficiency?

a. Albumin

b. Ferritin

c. Transferrin

d. Ceruloplasmin

e. Haptoglobin

Q35

During an acute inflammatory response, the concentration of certain plasma proteins increases. Which of the following is considered a positive "acute-phase reactant"?

a. Albumin

b. Transferrin

c. C-reactive protein (CRP)

d. Retinol-binding protein

e. Prealbumin

Q36

The immunoglobulin class that is the first to be secreted in a primary immune response and has a pentameric structure is:

a. IgG

b. IgA

c. IgM

d. IgE

e. IgD

Q37

Which immunoglobulin is found in high concentrations in secretions such as saliva, tears, and colostrum, providing mucosal immunity?

a. IgG

b. IgA

c. IgM

d. IgE

e. IgD

Q38

A patient’s plasma protein electrophoresis shows a massive spike in the gamma-globulin region. This finding is most characteristic of:

a. Liver cirrhosis

b. Nephrotic syndrome

c. Multiple myeloma

d. Iron deficiency anemia

e. Acute inflammation

Introduction to Lipids (Classification & Functions)

Q39

Lipids are organic molecules that are generally insoluble in water. Which of the following is classified as a "simple lipid"?

a. Phospholipid

b. Glycolipid

c. Triglyceride

d. Sphingomyelin

e. Lipoprotein

Q40

Which lipoprotein is responsible for carrying dietary (exogenous) triglycerides from the intestine to the peripheral tissues?

a. Chylomicron

b. VLDL

c. LDL

d. HDL

e. IDL

Q41

Which of the following is an "essential" fatty acid that cannot be synthesized by humans and must be obtained from the diet?

a. Palmitic acid

b. Stearic acid

c. Oleic acid

d. Linoleic acid

e. Arachidic acid

Q42

Cholesterol is a vital sterol that serves as a precursor for the synthesis of all of the following EXCEPT:

a. Bile acids

b. Vitamin D

c. Steroid hormones

d. Vitamin C

e. Cell membranes

Q43

Which class of lipids is the major component of the alveolar surfactant that prevents the lungs from collapsing?

a. Triglycerides

b. Cholesterol esters

c. Phospholipids (Lecithin)

d. Glycolipids

e. Waxes

Mixed Review & Clinical Correlation

Q44

The enzyme Lysyl Oxidase, which is critical for forming cross-links in collagen and elastin, requires which metal as a cofactor?

a. Iron

b. Zinc

c. Copper

d. Magnesium

e. Manganese

Q45

Kwashiorkor is a clinical condition resulting from a severe deficiency of:

a. Total calories

b. Dietary fat

c. Dietary protein

d. Vitamin D

e. Vitamin C

Q46

Which amino acid is the primary precursor for the inhibitory neurotransmitter GABA?

a. Glycine

b. Glutamate

c. Tyrosine

d. Tryptophan

e. Histidine

Q47

A patient with Gout has an accumulation of which substance in the joints?

a. Calcium oxalate

b. Urate crystals

c. Cholesterol

d. Bilirubin

e. Cystine

Q48

Which amino acid is involved in the synthesis of Heme?

a. Alanine

b. Glycine

c. Valine

d. Leucine

e. Serine

Q49

The "T" (Tense) state of hemoglobin is characterized by:

a. High oxygen affinity

b. Low oxygen affinity

c. Absence of 2,3-BPG

d. Predominance in the lungs

e. Rupture of salt bridges

Q50

Which enzyme is responsible for sealing nicks in the sugar-phosphate backbone of DNA during replication and repair?

a. DNA Helicase

b. DNA Primase

c. DNA Polymerase

d. DNA Ligase

e. DNA Topoisomerase

Answer Key: Set 1

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	e	**11**	a	**21**	b	**31**	b	**41**	d
2	a	**12**	c	**22**	c	**32**	b	**42**	d
3	b	**13**	c	**23**	c	**33**	d	**43**	c
4	b	**14**	d	**24**	d	**34**	c	**44**	c
5	d	**15**	e	**25**	b	**35**	c	**45**	c
6	a	**16**	d	**26**	c	**36**	c	**46**	b
7	a	**17**	b	**27**	d	**37**	b	**47**	b

8	e	**18**	b	**28**	e	**38**	c	**48**	b
9	a	**19**	c	**29**	d	**39**	c	**49**	b
10	d	**20**	b	**30**	c	**40**	a	**50**	d